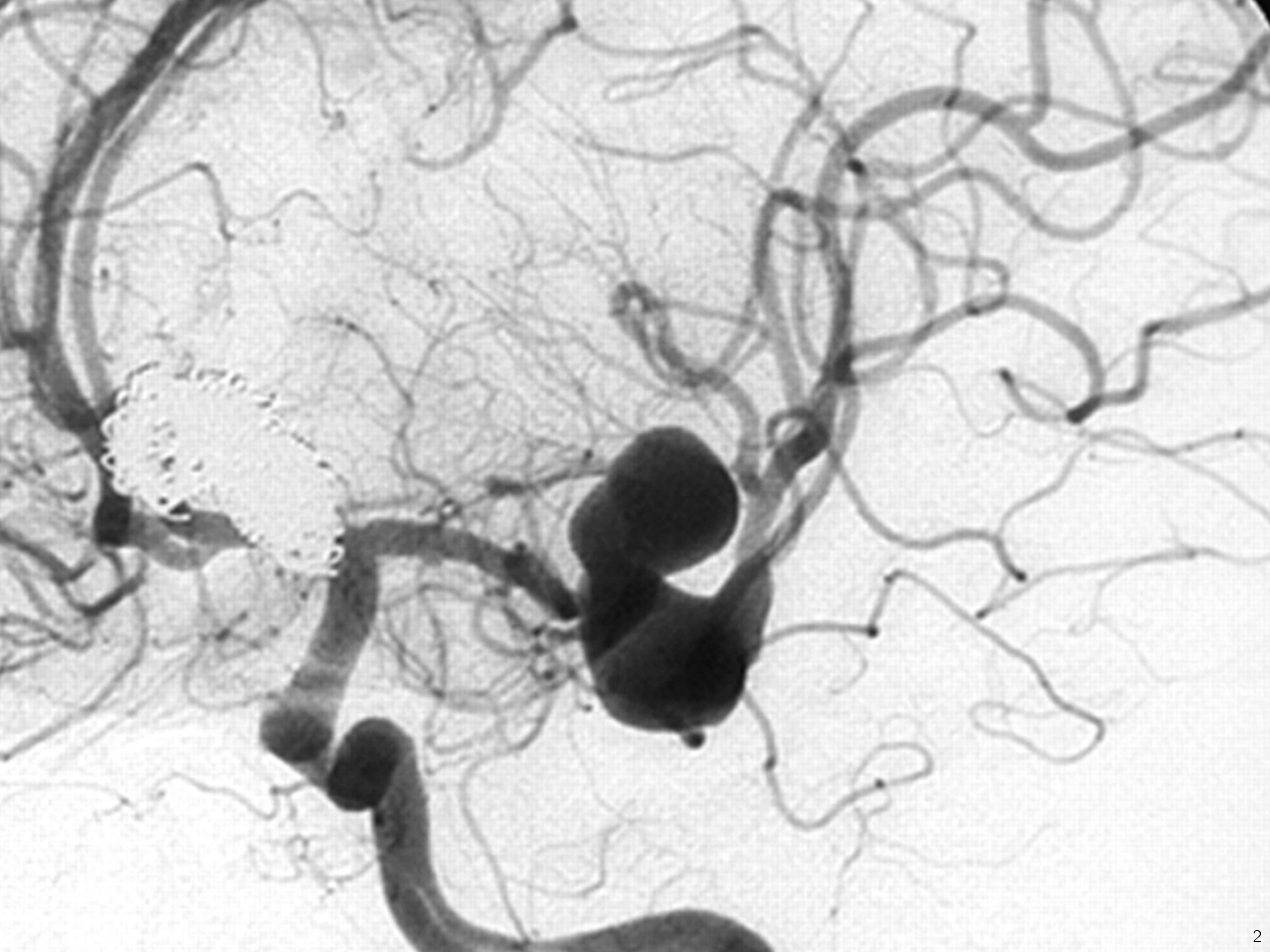


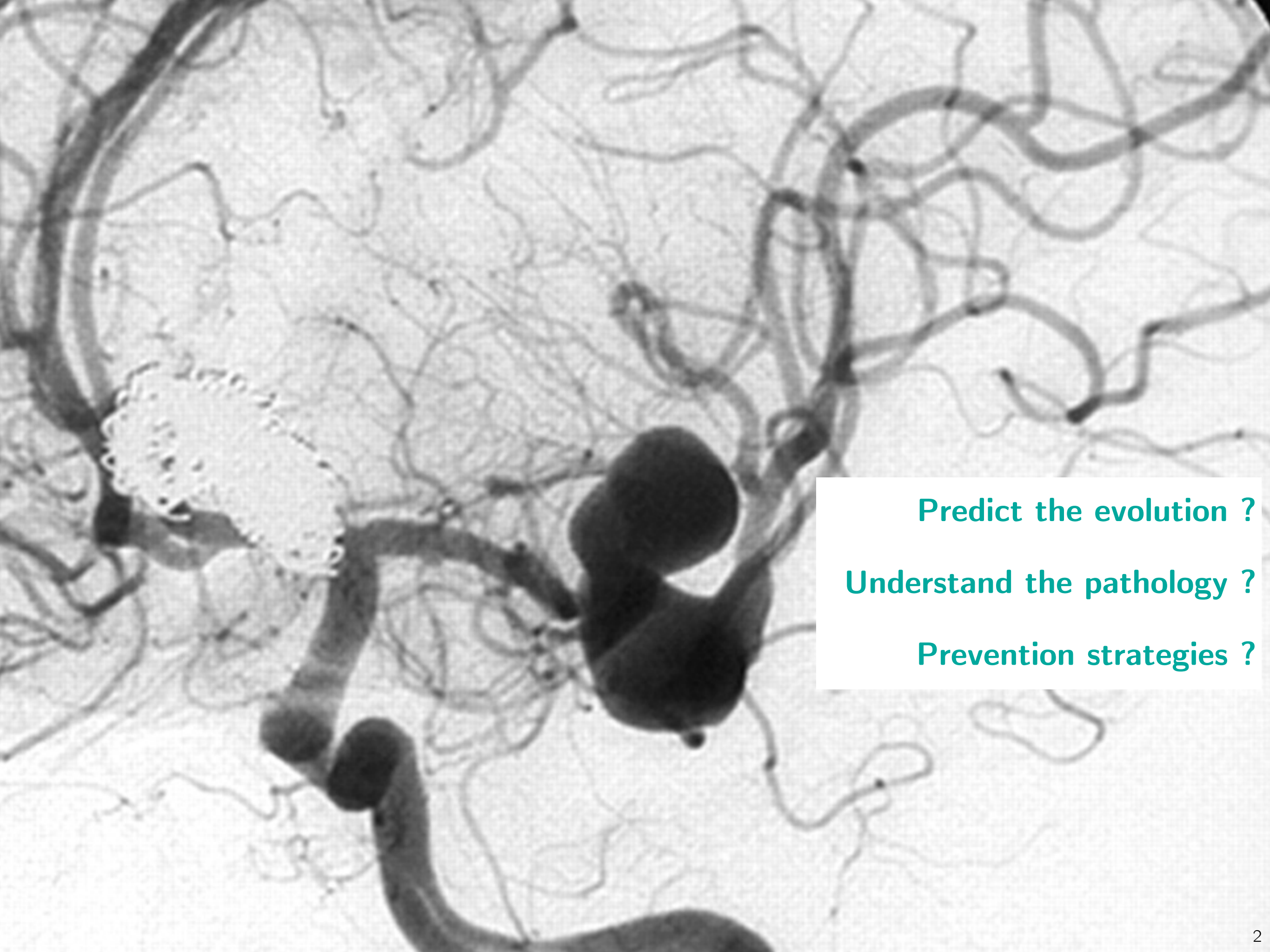
Provenance as a key material to improve FAIR-ness in data-driven (life) sciences

Alban Gaignard, PhD, CNRS

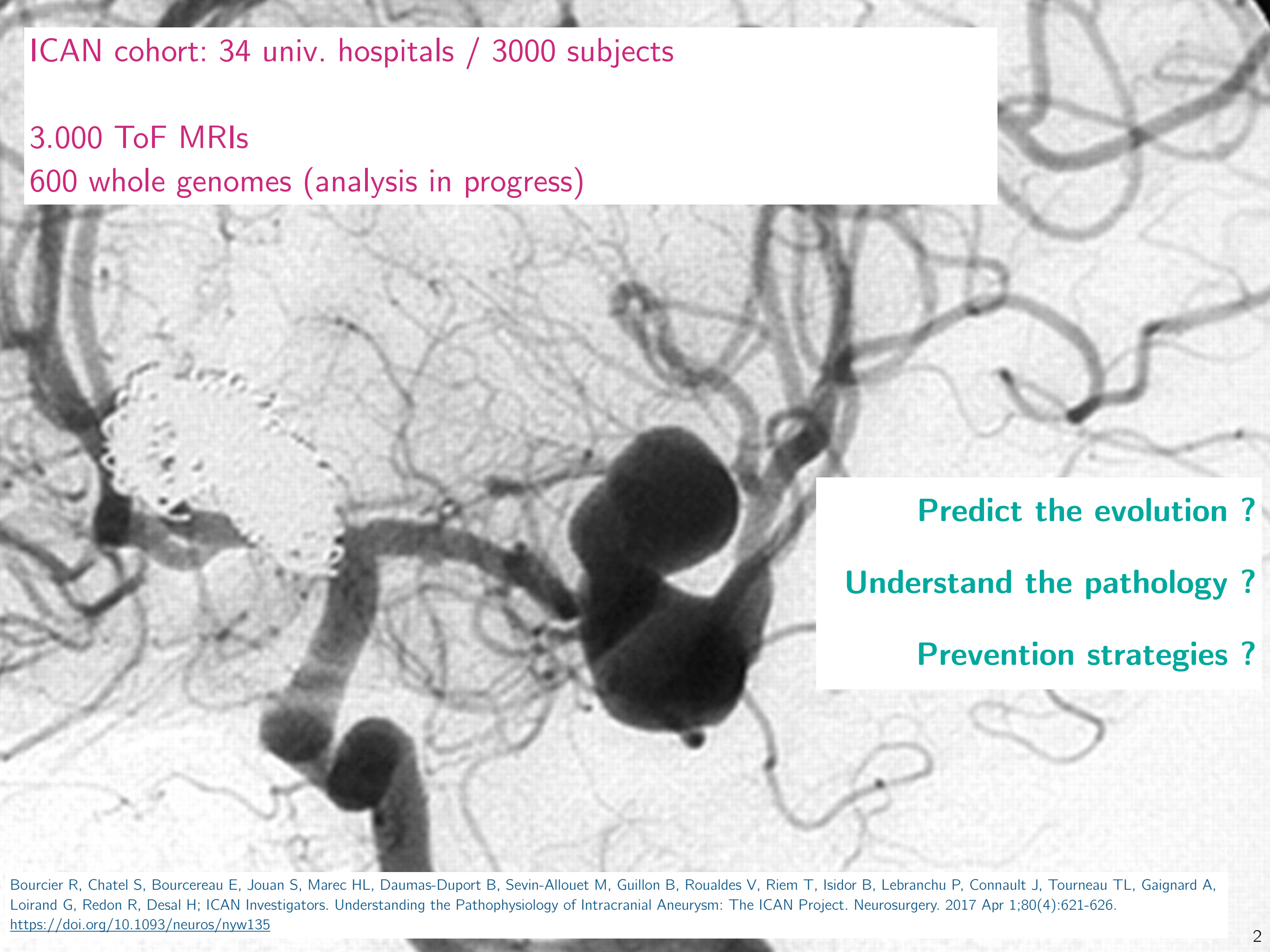
Colloque "Réplicabilité et Reproductibilité de la Recherche"
MITI, CNRS

Paris, 08 septembre 2023





Predict the evolution ?
Understand the pathology ?
Prevention strategies ?

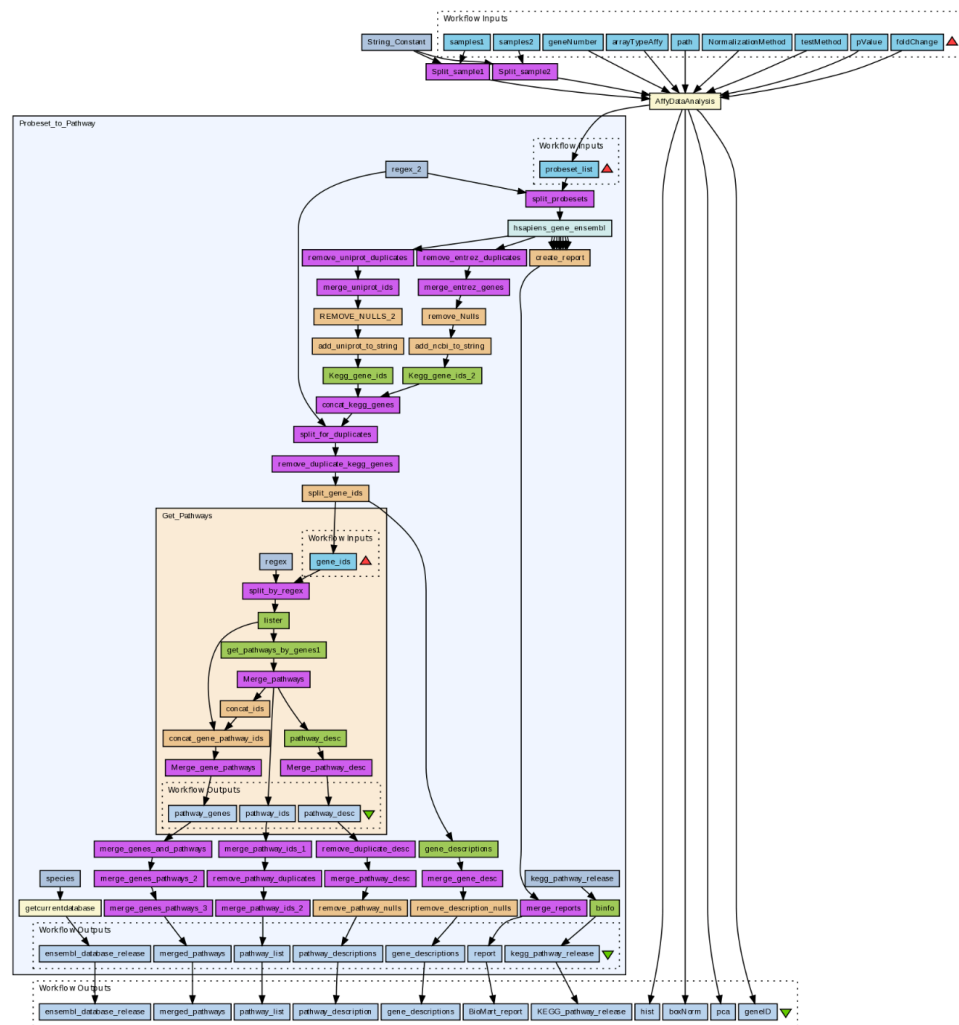


ICAN cohort: 34 univ. hospitals / 3000 subjects

3.000 ToF MRIs

600 whole genomes (analysis in progress)

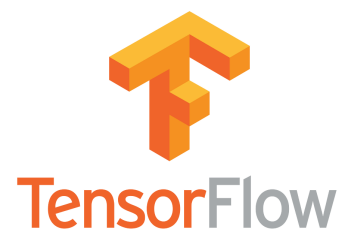
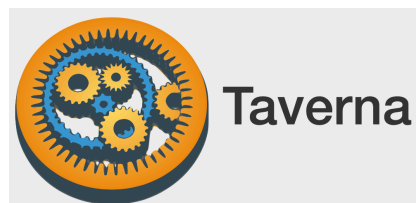
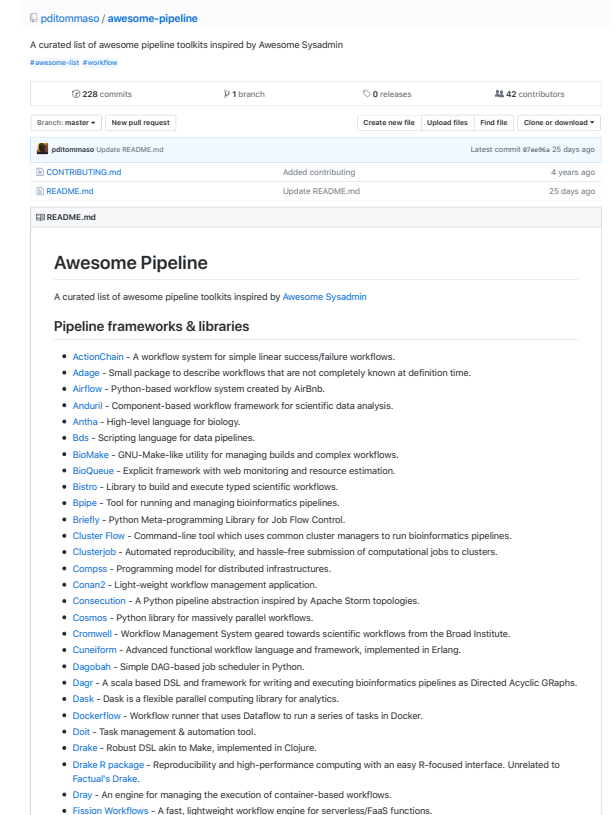
Predict the evolution ?
Understand the pathology ?
Prevention strategies ?



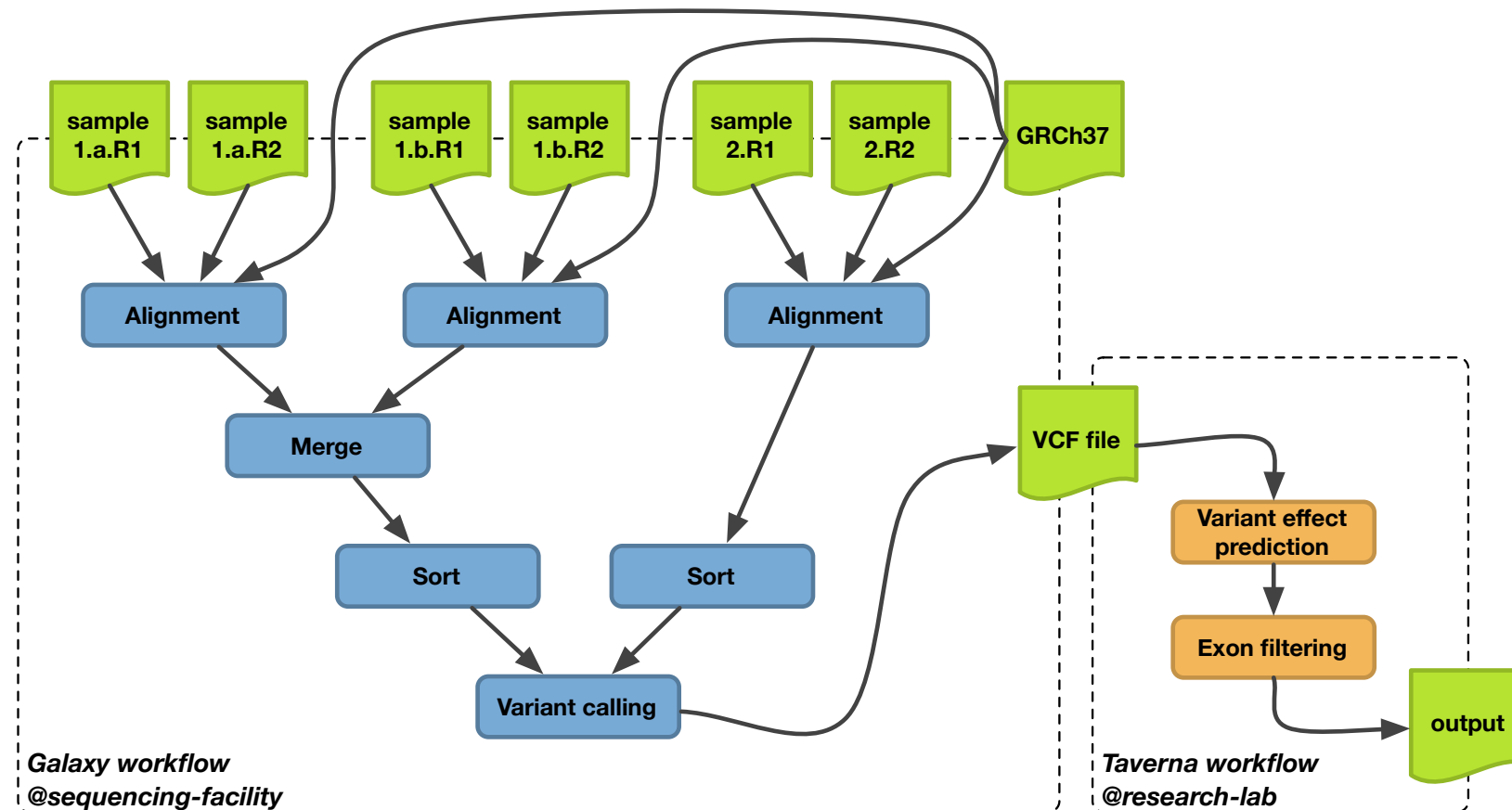
Workflows to enhance trust in scientific results :

- **automation** (scalability)
- **abstraction** (methods sharing)
- **traceability** (provenance)

nextflow



Interpreting bioinformatics analysis results ?



« Which alignment algorithm was used when predicting this pathogenic score ? »



« A new version of a reference genome is available, which genome was used when predicting these phenotypes ? »

Need for an overall tracking of provenance over multiple workflows !

Provenance in Computer Science

« Provenance information describes the **origins** and the **history** of data in its **life cycle**. »

« Today, data is often made **available on the Internet** with **no centralized control** over its integrity: data is constantly being **created**, **copied**, **moved** around, and **combined** indiscriminately. Because information sources (or different parts of a single large source) may vary widely in terms of quality, it is essential to provide **provenance** and other **context** information which can **help end users judge** whether query **results are trustworthy**. »

20 years of provenance ...

Foundations and Trends® in
Databases
Vol. 1, No. 4 (2007) 379–474
© 2009 J. Cheney, L. Chiticariu and W.-C. Tan
DOI: 10.1561/1900000006



Provenance in Databases: Why, How, and Where

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and Wang-Chiew Tan³

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chiti@almaden.ibm.com

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Abstract

Different notions of provenance for database queries have been proposed and studied in the past few years. In this article, we detail three main notions of database provenance, some of their applications, and compare and contrast amongst them. Specifically, we review why, how, and where provenance, describe the relationships among these notions of provenance, and describe some of their applications in confidence computation, view maintenance and update, debugging, and annotation propagation.



W3C Recommendation



PROV-O: The PROV Ontology

W3C Recommendation 30 April 2013

This version:

<http://www.w3.org/TR/2013/REC-prov-o-20130430/>

Latest published version:

<http://www.w3.org/TR/prov-o/>

Implementation report:

<http://www.w3.org/TR/2013/NOTE-prov-implementations-20130430/>

Previous version:

<http://www.w3.org/TR/2013/PR-prov-o-20130312/>

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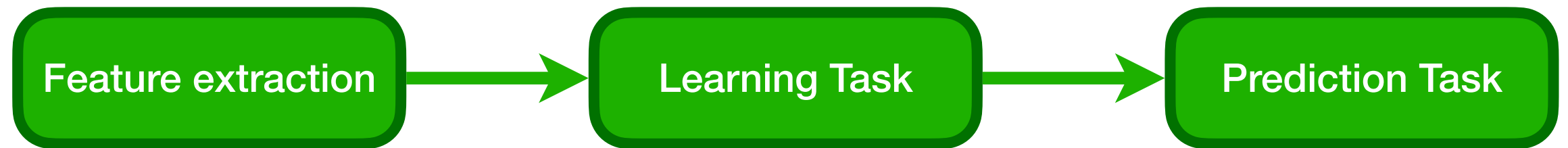
[Stephan Zednik](#), Rensselaer Polytechnic Institute, USA

[Jun Zhao](#), University of Oxford, UK

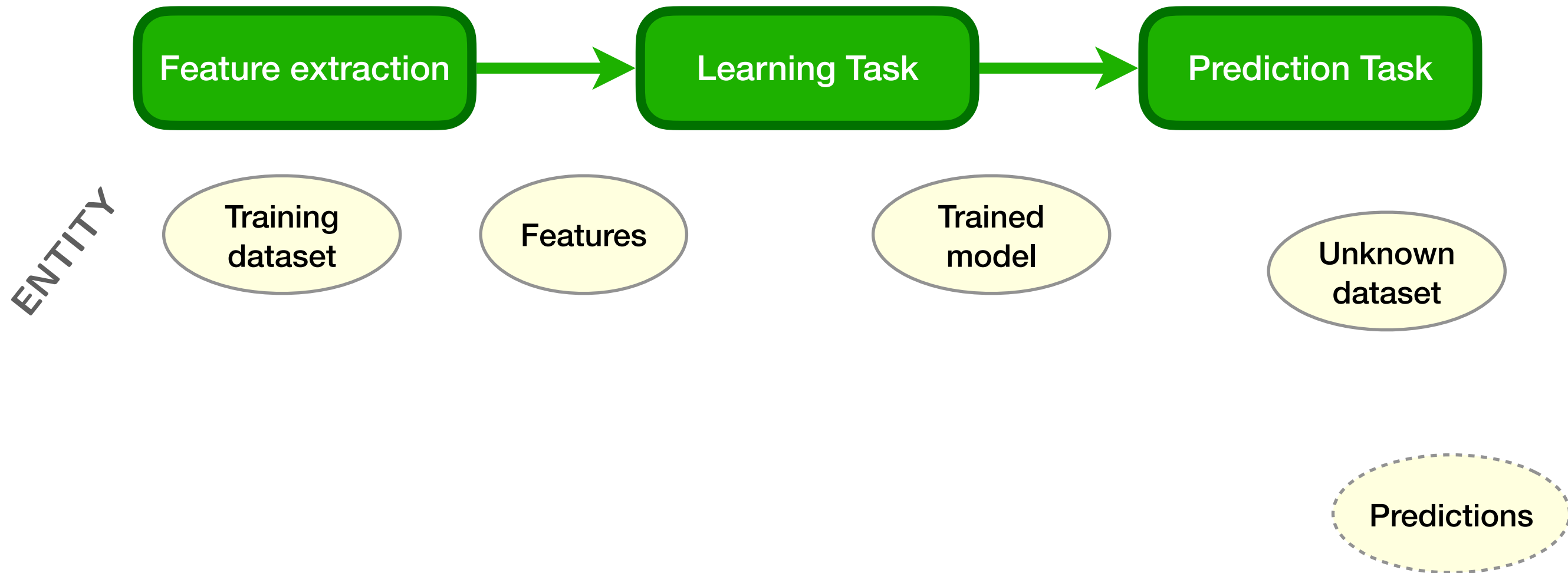
Which DB tuples
contribute to a query
result ? Which operators ?

A standardized lightweight,
extensible model to represent
Provenance on the Web.

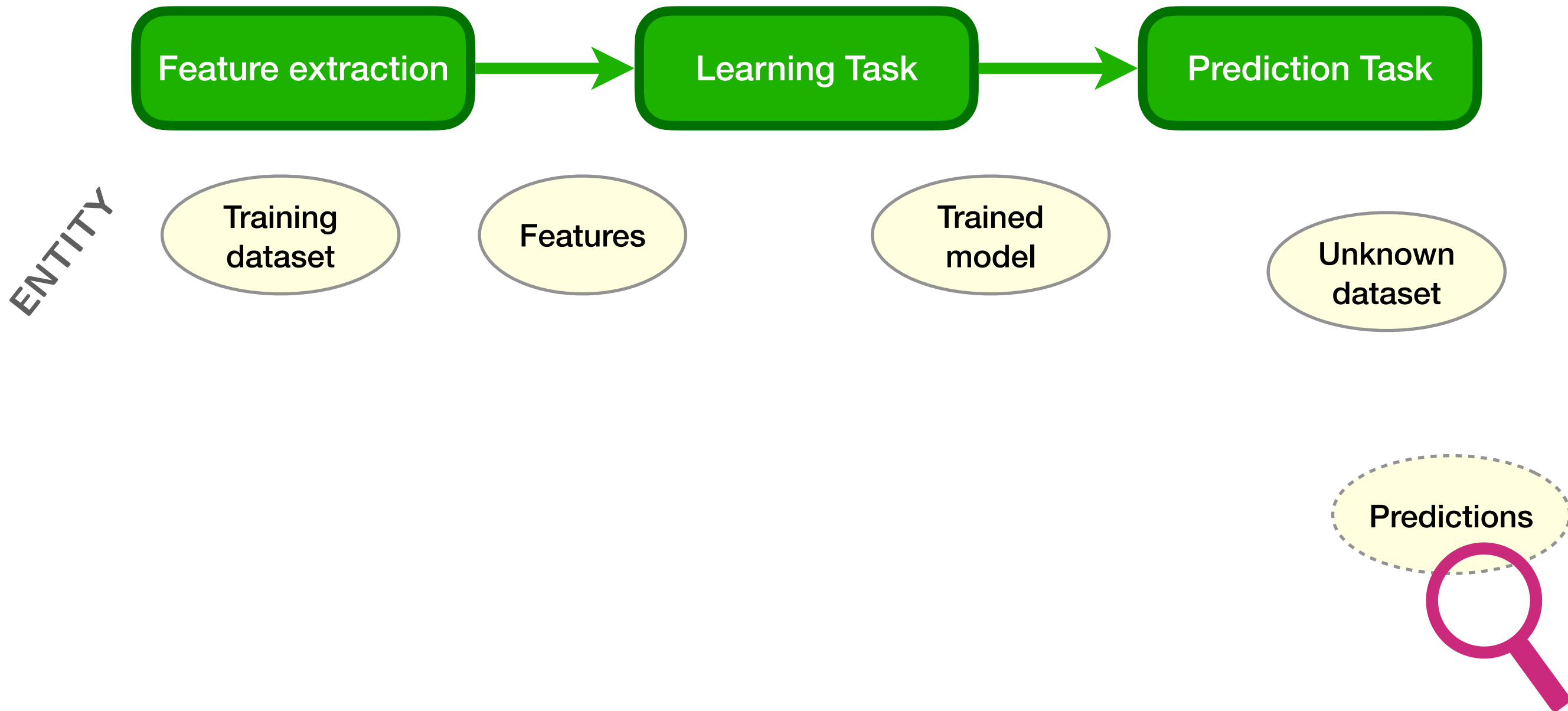
Data / Algorithms / Researchers



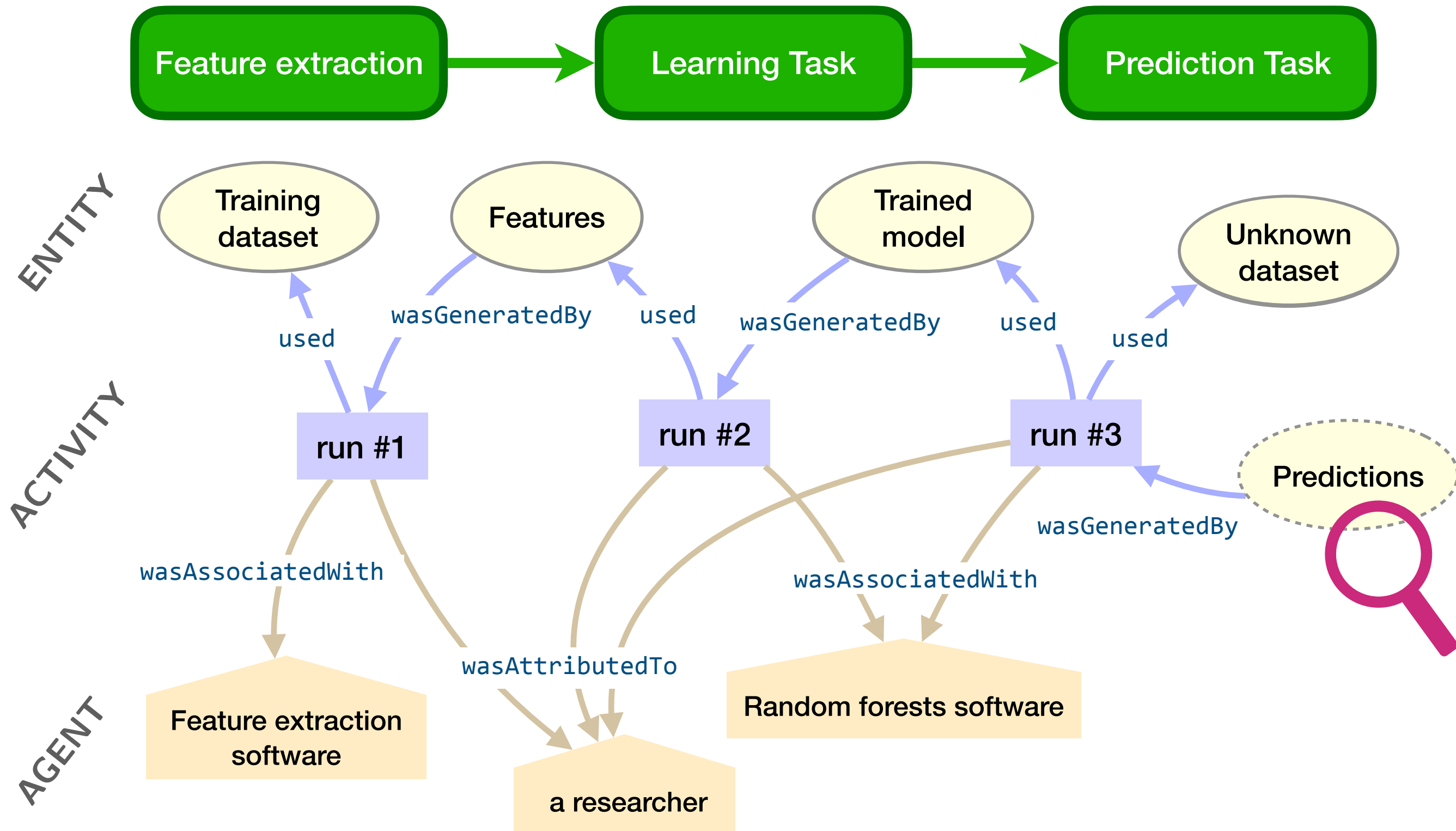
Data / Algorithms / Researchers



Data / Algorithms / Researchers



Data / Algorithms / Researchers



PROV extensions

[nature](#) > [scientific data](#) > [articles](#) > [article](#)

[Open Access](#) | [Published: 06 December 2016](#)

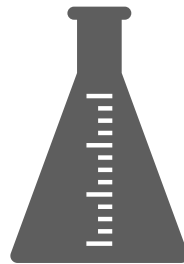
Sharing brain mapping statistical results with the neuroimaging data model

[Camille Maumet](#) , [Tibor Auer](#), [Alexander Bowring](#), [Gang Chen](#), [Samir Das](#), [Guillaume Flandin](#), [Satrajit Ghosh](#), [Tristan Glatard](#), [Krzysztof J. Gorgolewski](#), [Karl G. Helmer](#), [Mark Jenkinson](#), [David B. Keator](#), [B. Nolan Nichols](#), [Jean-Baptiste Poline](#), [Richard Reynolds](#), [Vanessa Sochat](#), [Jessica Turner](#) & [Thomas E. Nichols](#)

[Scientific Data](#) **3**, Article number: 160102 (2016) | [Cite this article](#)

5277 Accesses | 30 Citations | 42 Altmetric | [Metrics](#)

<https://doi.org/10.1038/sdata.2016.102>



Research | [Open Access](#) | [Published: 31 January 2022](#)

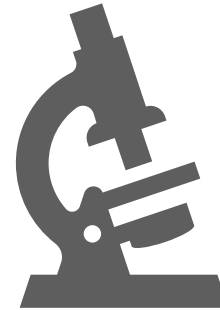
Structure-based knowledge acquisition from electronic lab notebooks for research data provenance documentation

[Max Schröder](#) , [Susanne Staehle](#), [Paul Groth](#), [J. Barbara Nebe](#), [Sascha Spors](#) & [Frank Krüger](#)

[Journal of Biomedical Semantics](#) **13**, Article number: 4 (2022) | [Cite this article](#)

4265 Accesses | 6 Citations | 9 Altmetric | [Metrics](#)

<https://doi.org/10.1186/s13326-021-00257-x>



Research | [Open Access](#) | [Published: 06 January 2022](#)

End-to-End provenance representation for the understandability and reproducibility of scientific experiments using a semantic approach

[Sheeba Samuel](#)  & [Birgitta König-Ries](#)

[Journal of Biomedical Semantics](#) **13**, Article number: 1 (2022) | [Cite this article](#)

3475 Accesses | 3 Citations | 8 Altmetric | [Metrics](#)

<https://doi.org/10.1186/s13326-021-00253-1>



Astronomical Pipeline Provenance: A Use Case Evaluation

Authors:

Michael A. C. Johnson, *Institute of Data Science (DLR) and Max Planck Institute for Radio Astronomy*; Marcus Paradies and Marta Dembska, *Institute of Data Science (DLR)*; Kristen Lackeos, Hans-Rainer Klöckner, and David J. Champion, *Max Planck Institute for Radio Astronomy*; Sirko Schindler, *Institute of Data Science (DLR)*

<https://doi.org/10.48550/arXiv.2109.10759>

[Home](#) > [Provenance and Annotation of Data and Processes](#) > [Conference paper](#)

Towards a Provenance Management System for Astronomical Observatories

[Mathieu Servillat](#) , [François Bonnarel](#), [Catherine Boisson](#), [Mireille Louys](#), [Jose Enrique Ruiz](#) & [Michèle Sanguillon](#)

Conference paper | [First Online: 09 July 2021](#)

515 Accesses | 1 Citations | 8 Altmetric

Part of the [Lecture Notes in Computer Science](#) book series (LNISA, volume 12839)

https://doi.org/10.1007/978-3-030-80960-7_20

Many expectations ...

Comparability, transparency, confidence +

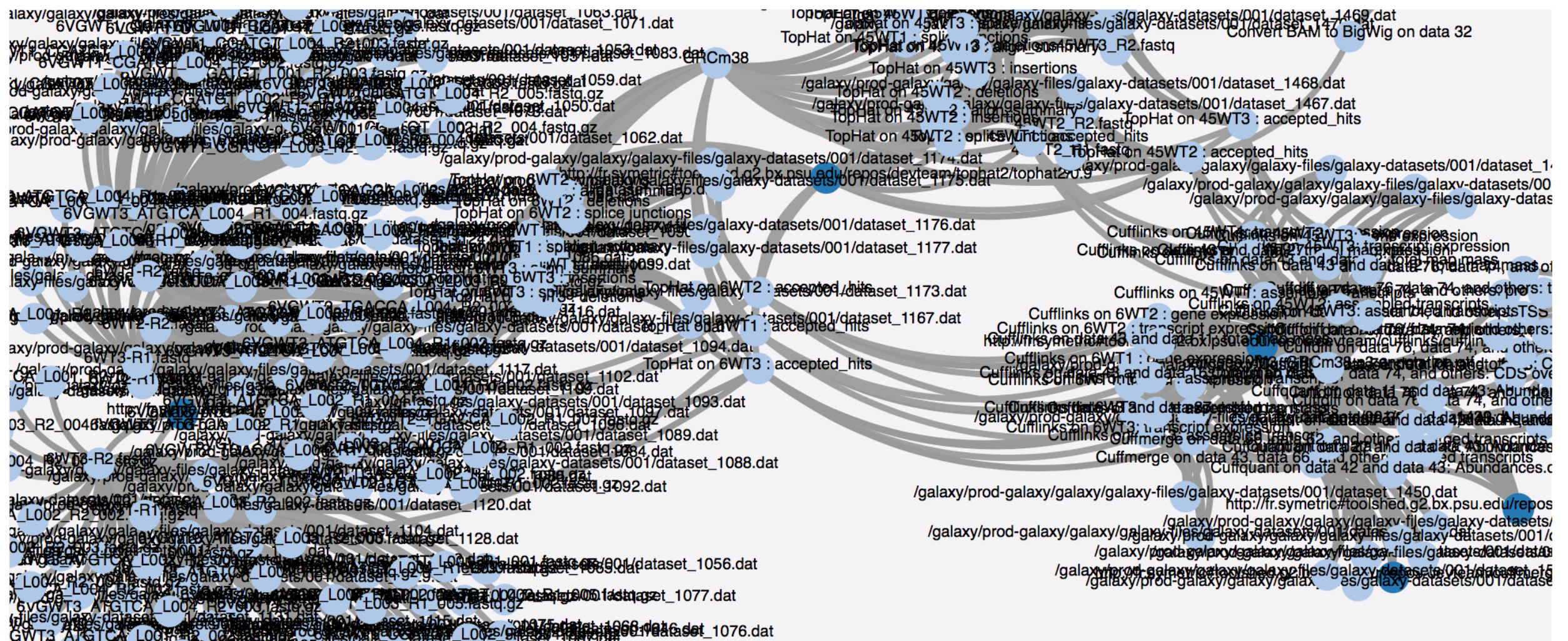
- ▶ **Citing** researchers and organisations
- ▶ Identifying **critical** data / software **resources** associated to scientific results
- ▶ Identifying possible **bias** when reusing / sharing pre-trained models

Reuse instead of
re-execution ?

Is PROV enough for reuse ?

```
11 a prov:Bundle, prov:Entity;  
12 prov:wasAttributedTo <#galaxy2prov>;  
13 prov:generatedAtTime "2016-04-14T18:18:37.000409"^^xsd:dateTime;  
14 .  
15  
16 <#72486b583fe152f0>  
17 a prov:Activity ;  
18 prov:wasAssociatedWith <#cat1> ;  
19 prov:startedAtTime "2015-12-15T12:54:50.749845"^^xsd:dateTime;  
20 prov:endedAtTime "2015-12-15T12:55:57.016799"^^xsd:dateTime.
```

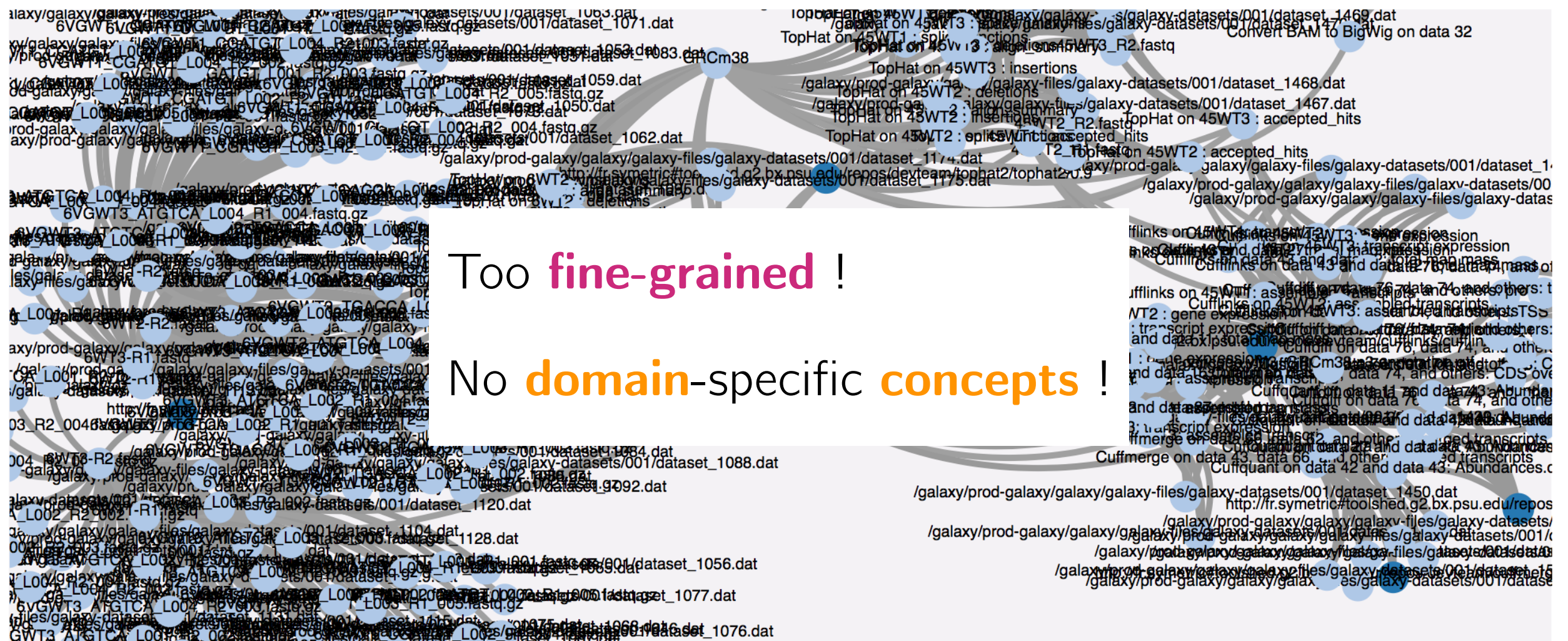
Visualise



Is PROV enough for reuse ?

```
11 a prov:Bundle, prov:Entity;  
12 prov:wasAttributedTo <#galaxy2prov>;  
13 prov:generatedAtTime "2016-04-14T18:18:37.000409"^^xsd:dateTime;  
14 .  
15  
16 <#72486b583fe152f0>  
17 a prov:Activity ;  
18 prov:wasAssociatedWith <#cat1> ;  
19 prov:startedAtTime "2015-12-15T12:54:50.749845"^^xsd:dateTime;  
20 prov:endedAtTime "2015-12-15T12:55:57.016799"^^xsd:dateTime;
```

Visualise



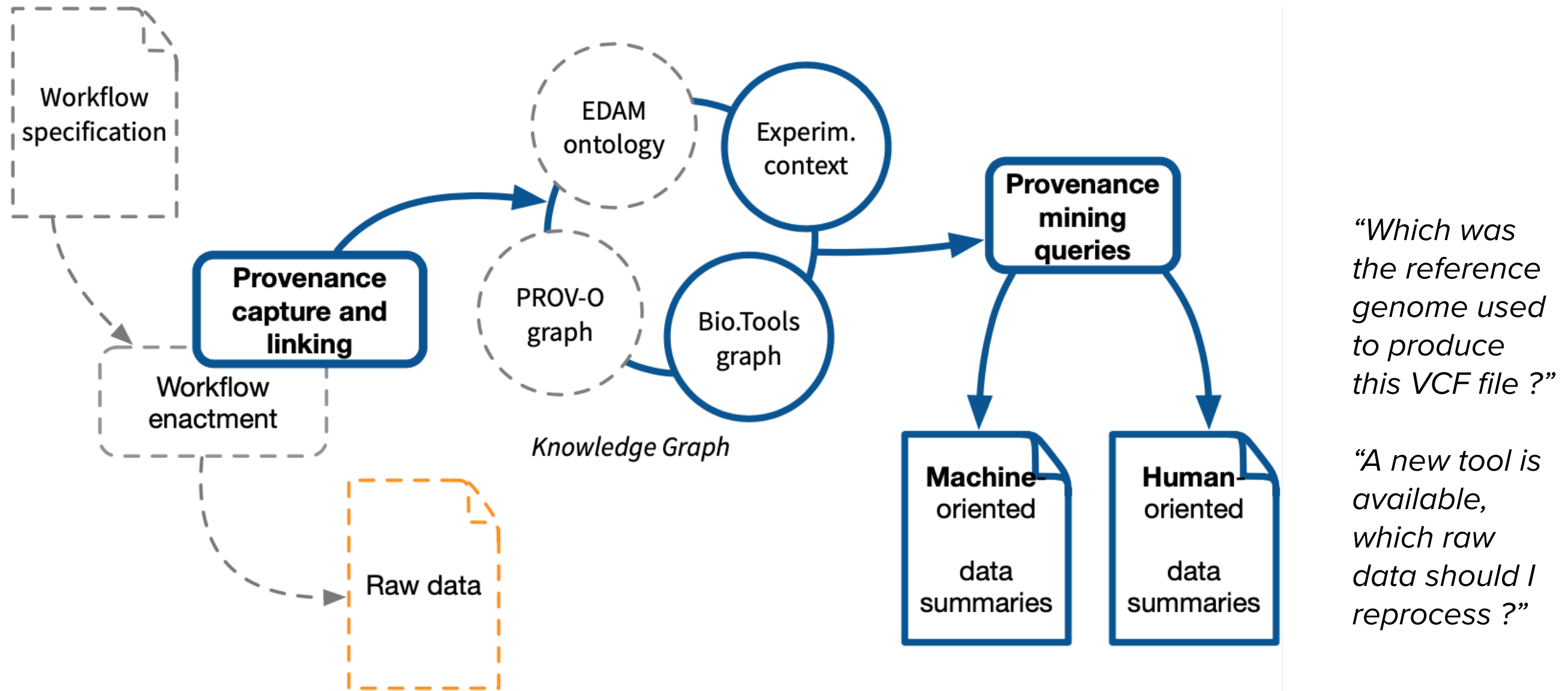
Annotated bioinformatics tools catalog

Ontology terms (EDAM) to "tag"

- ▶ topic
- ▶ input data
- ▶ processing
- ▶ output data

The screenshot displays the JASPAR tool page on the Elixir bio.tools catalog. The header includes the Elixir logo, a search bar, and navigation links for 'About', 'Login', and 'Sign-up'. The JASPAR tool is identified with a verified ID and its URL. Below this, a series of tags categorizes the tool: 'Transcription factors and regulatory sites', 'Gene regulation', 'Genomics', 'Human biology', 'Plant biology', and 'Model organisms'. Further tags include 'Mature', 'CC-BY-NC-4.0', 'Free of charge', 'Open access', 'Web API', 'Web application', 'Database portal', and 'Python'. A central diagram illustrates the tool's workflow: input fields for transcription factor details (name, identifier, taxon, family, species, UniProt ID, text) and DNA sequence (FASTA) lead to a 'Database search' and 'Transcription factor binding site prediction' process, resulting in 'JASPAR profile ID' output. The bottom section provides 'Credits & Support' (listing primary contacts Albin Sandelin, Boris Lenhard, Wyeth Wasserman, and Anthony Mathelier), 'Documentation' (links to docs, FAQ, and API), 'Downloads' (downloads page), and 'Links' (Twitter, GitHub repository). 'Publication details' at the bottom show 38 citations and 586 downloads for the primary publication: 'JASPAR 2018: Update of the open-access database of transcription factor binding profiles and its web framework' by Khan et al.

Approach

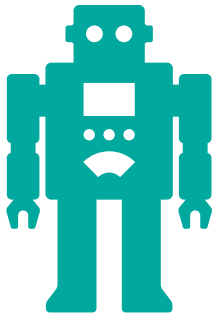


Methods and tools

graph pattern matching, inference rules, SPARQL, Python, Jupyter notebooks

Machine-Human readable experiment summaries

Machine-Human readable experiment summaries



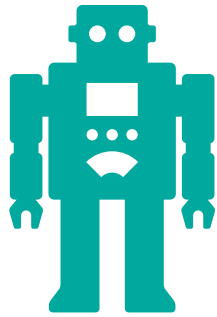
```
[...]
:head {
  _:np1 a np:Nanopublication .
  _:np1 np:hasAssertion :assertion .
  _:np1 np:hasProvenance :provenance .
  _:np1 np:hasPublicationInfo :pubInfo .
}

:assertion {
  <http://snakemake-provenance/Samples/Sample1/
  BAM/Sample1.merged.bai> rdfs:seeAlso
  <http://edamontology.org/operation_3197> .

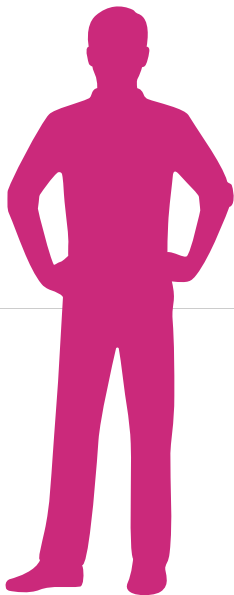
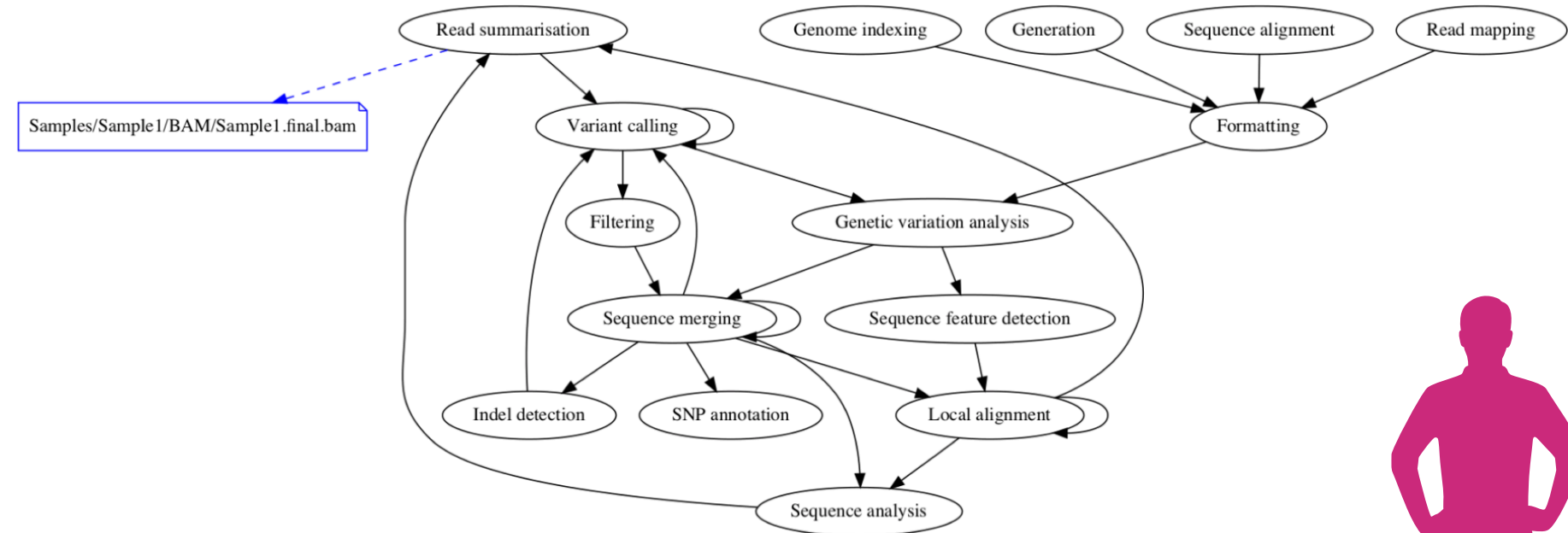
  <http://snakemake-provenance/VCF/hapcaller.
  indel.recal.filter.vcf.gz> rdfs:seeAlso
  <http://edamontology.org/operation_3695> .
}
[...]
```

1. It's possible to automatically
annotate produced data with
ontology terms

Machine-Human readable experiment summaries



```
[...]  
:head {  
  _:np1 a np:Nanopublication .  
  _:np1 np:hasAssertion :assertion .  
  _:np1 np:hasProvenance :provenance .  
  _:np1 np:hasPublicationInfo :pubInfo .  
}  
  
:assertion {  
  <http://snakemake-provenance/Samples/Sample1/  
  BAM/Sample1.merged.bai> rdfs:seeAlso  
  <http://edamontology.org/operation_3197> .  
  
  <http://snakemake-provenance/VCF/hapcaller.  
  indel.recal.filter.vcf.gz> rdfs:seeAlso  
  <http://edamontology.org/operation_3695> .  
}  
[...]
```



```
...  
The file Samples/Sample1/BAM/Sample1.realign.bai results from  
tool gatk2_indel_realigner-IP which Locally align two or more molecular  
sequences.  
  
It was produced in the context of Rare Coding Variants in ANGPTL6 Are  
Associated with Familial Forms of Intracranial Aneurysm  
...
```

1. It's possible to automatically
annotate produced data with
ontology terms

2. It's possible to automatically display the **typical bioinformatics tasks** data originate from
3. It's possible to document data with **text** leveraging ontology definitions (EDAM)

Wrap-up

Take-home message & perspectives

- ▶ PROV = **consensual model** to represent and share provenance on the **web**
- ▶ $\neq$ communities/tools $\rightarrow$ PROV **heterogeneity**
- ▶ Computational **reproducibility**
 - $\rightarrow$ fine-grained capture (hardware ? OS ?)
 - $\rightarrow$ information overload for humans
- ▶ FAIR
 - ⚠ **R1.2** criteria poorly validated¹
 - ⚠ domain-specific annotations
 - $\rightarrow$ scientific context ? DMPs ?

—— Future works ——
PEPR Santé-Numérique
(ShareFAIR, NeuroVasc)



¹ FAIR-Checker: supporting digital resource findability and reuse with Knowledge Graphs and Semantic Web standards. J Biomed Semant 14, 7 (2023). <https://doi.org/10.1186/s13326-023-00289-5>

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Khalid Belhajjame,
LAMSAD, University of
Paris-Dauphine, PSL

GDR **MaDICS**
action **ReproVirtuFlow**